

VIKAS RAJPUT

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PROFILE

BS Data Science fresher with hands-on experience building production AI systems — multi-agent LLM pipelines, Retrieval-Augmented Generation (RAG), event-driven agent orchestration, and FastAPI-based deployment. Built and deployed 5 end-to-end AI engineering projects covering LangGraph, LangChain, PyTorch, vector databases, and agentic AI with guardrails. Seeking an AI Engineering internship to contribute to real-world intelligent system development.

AI ENGINEERING PROJECTS

[AgentOps-Engine](#) | [deployment in progress](#)

Designed and built an event-driven micro-agent hierarchy modelled on production ERP maritime workflows — BookingCreated events trigger 4 parallel micro-agents (Validation, Compliance, Billing, Follow-Up) coordinated by a Screen Manager, Implemented a 3-level agent architecture: Domain Director → Screen Manager → Micro-Agents, each with Pydantic input validation, confidence thresholds, and human escalation triggers

Integrated a Neo4j-ready knowledge graph (Booking → Customer → Vessel) with an in-memory mock that shares an identical interface; agents query entity context before making decisions , Built a Redis pub/sub event bus abstraction; in-memory and Redis modes are swappable via environment variable with zero code changes , Wired LangSmith distributed tracing via TracingMiddleware; every agent action logged to an immutable audit trail with rollback flags, decision reasons, and latency metrics,

stack : FastAPI · LangSmith · Redis · Neo4j · asyncio · Pydantic

[ResearchMind](#) · [Live demo](#)

Stack: LangGraph · Tavily Search API · LangSmith · FastAPI · Streamlit · MemorySaver · Groq

Built a production-grade multi-agent system using LangGraph StateGraph with 3 specialist agents (Researcher → Analyst → Writer) and a supervisor with conditional edge routing, Implemented per-session memory using LangGraph MemorySaver checkpointer — agent state persists across queries per thread_id; swappable with SqliteSaver or PostgresSaver for production, Designed guardrails on every agent output: confidence scoring, human review flag, structured Pydantic FinalReport schema, and error fallback routing that prevents silent failures

Integrated DuckDuckGo search for real-time factual grounding in the Researcher agent; LangSmith tracing captures full trace tree including LLM calls, tool calls, and state transitions

Deployed FastAPI backend with typed request/response schemas; Streamlit frontend displays citation sources, confidence score, iteration count, and per-agent latency

[DocuMind](#) · [live demo](#)

Stack: LangChain · HuggingFace sentence-transformers · Gemini · Streamlit, · FAISS · ChromaDB · RAGAS · LangSmith · FastAPI · Groq

Built an end-to-end document QA pipeline: PDF/TXT ingestion → recursive chunking → dual indexing (BM25 sparse + FAISS dense) → Reciprocal Rank Fusion (RRF) → cross-encoder reranking → citation-grounded LLM answer Implemented Reciprocal Rank Fusion using formula $RRF(d) = \sum 1/(k + rank_i(d))$ to merge BM25 and dense retrieval rankings without score normalization, improving retrieval quality over either method alone Added RAGAS evaluation pipeline measuring faithfulness, ROUGE-L, and answer relevancy over a Q&A test set; results persisted to JSON for reproducibility and reporting , Instrumented all retrieval stages with LangSmith @traceable decorators (hybrid_retrieval, rrf_fusion, rerank, rag_generate) enabling per-step latency and token-cost visibility, Served streaming responses via FastAPI StreamingResponse with model pre-loaded at startup via lifespan hook; Streamlit UI provides chat interface, document manager, and evaluation dashboard

MACHINE LEARNING & DEEP LEARNING PROJECTS

Audio Genre Classification — AST Fine-tuning [deployed on streamlit](#)

Stack: *PyTorch · Audio Spectrogram Transformer (AST) · CRNN · HuggingFace Transformers · LoRA · HuggingFace Transformers · FastAPI · Streamlit*

- Build a Genre classification model by Fine-tuned Audio Spectrogram Transformer on audio dataset; combined with CRNN architecture for hybrid spectrogram + sequential modelling
- Deployed on HuggingFace Spaces — public live inference demo; model card published at huggingface.co/Vikas25S

Comment Category Prediction — NLP Classification [kaggle](#)

Stack: *Python · Scikit-learn · TF-IDF · Ensemble Methods · NLP*

- Multi-class text classifier for comment categorisation using TF-IDF feature engineering + ensemble ML models (XGBoost / LightGBM)
- Applied k-fold cross-validation and hyperparameter tuning; evaluated on macro F1 score across imbalanced class distribution
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TECHNICAL SKILLS

LLM / AI	LangChain, LangGraph, LangSmith, Retrieval-Augmented Generation (RAG), Agentic AI, Prompt Engineering, Fine-tuning (LoRA / QLoRA), Multi-Agent Systems.
Frameworks	FastAPI, Streamlit, PyTorch, Scikit-learn, HuggingFace Transformers, Sentence-Transformers
Deployment	Docker, Docker Compose, Railway, Streamlit Cloud, Hugging Face Spaces, GitHub Actions, CI/CD
Databases / Search	ChromaDB, FAISS, Redis, SQLite, Vector Search, Cosine Similarity, Max Marginal Relevance (MMR), Hybrid BM25 Search
Languages	Python (advanced), SQL, HTML/CSS, JavaScript (basic)
ML / DL	Transformers, CNNs, XGBoost, LightGBM, Time-Series Forecasting, NLP, Audio Classification, Model Evaluation
Tools	Git, GitHub, Kaggle, Jupyter, Pytest, Pydantic, Pandas, NumPy, LangSmith, MLflow

EDUCATION

BS Data Science and Applications *Expected 2026*

Indian Institute of Technology Madras (IIT Madras)

Relevant Coursework: *Machine Learning, Deep Learning & GenAI, Statistics, Linear Algebra, Tools in Data Science, Database Management*

ONLINE PROFILES

GitHub: github.com/23f3001800 • **HuggingFace:** huggingface.co/Vikas25S • **Kaggle:** kaggle.com/vikas25s

ADDITIONAL

All 5 projects are publicly available on GitHub with full documentation and deployed live demo URLs

Familiar with LoRA / QLoRA fine-tuning concepts for domain-adapted LLMs; actively building toward production fine-tuning workflows

Available to start immediately for internship